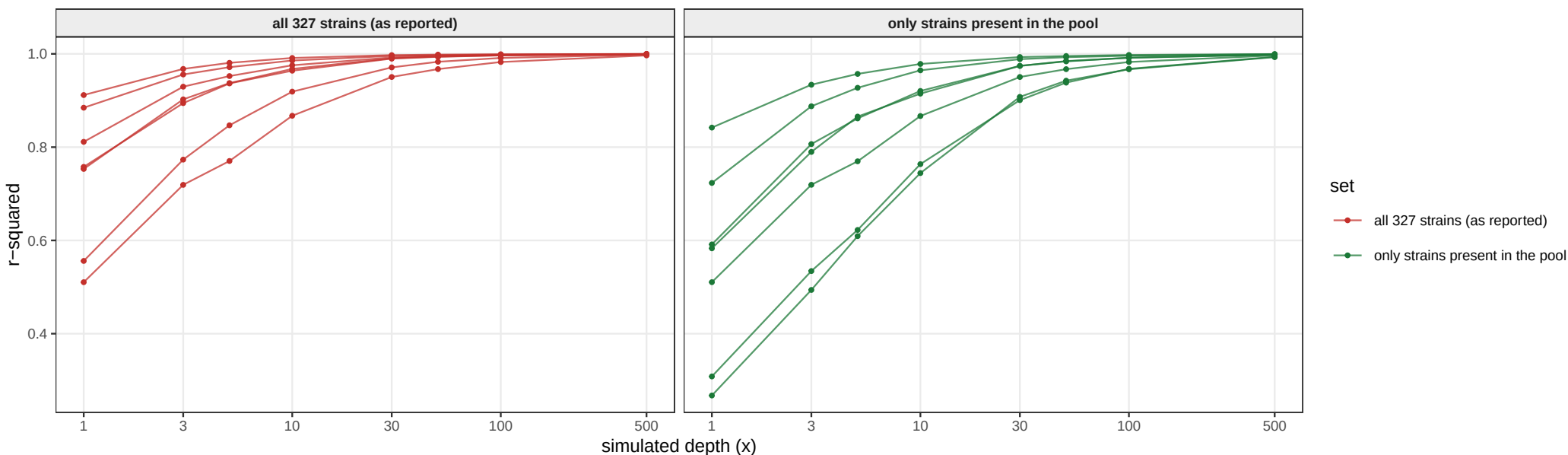


Part A: the absent strains do inflate it, by 0.07 at 10x and 0.19 at 1x

One line per trait. Absent strains have fitness exactly 0 and NNLS returns exactly 0 for them, zero being a hard boundary of the feasible set. Dropping them takes the mean from 0.953 to 0.879 at 10x and from 0.741 to 0.547 at 1x — real, but not the whole story, and not 'most of the recovery'.



The prediction failed, and the real driver is dispersion

I predicted the drop would track how MANY strains are absent. It does not (left). It tracks how spread out the present strains are (right): a tight cluster has little variance of its own, so the distant zeros supply the range and removing them collapses the fit. PC1 is 74.6% absent and loses 0.026.

Part B: privateness is no more predictive in a perfect world

The simulation solves with the same matrix it generated from, so the only error is binomial sampling. Privateness still explains only adj R2 0.21 here against 0.165 on real MIP-seq. The variance it leaves is not reference quality — adding abundance takes the clean-case model from 0.21 to 0.47.

